



Foundational Rollup Architecture IP and BlackRock's To- kenization Roadmap

Zach Kelling

Lux Industries

2026

Executive summary

BlackRock’s institutional tokenization roadmap — USD Institutional Digital Liquidity Fund (BUIDL), iShares Bitcoin Trust (IBIT) custody operations, the announced expansion of tokenized funds across additional asset classes, and the future migration of these holdings to dedicated rollup infrastructure — depends on a class of blockchain transaction-control architecture covered by foundational intellectual property held by Lux Industries, Inc.

The principal asset is U.S. Patent 11,803,853 B2 (“Controlling transactions on a network”), originally issued to Nahmii AS of Norway in October 2023 and acquired by Lux Industries in late 2025. The patent claims an architecture in which an off-chain operator originates transaction records, an independent set of validating nodes determines completeness and correctness, and a smart-contract execution agent on a parent blockchain permits or denies execution based on the validation result. This architecture is the structural foundation of every modern Layer-2 rollup, every Alt-DA data-availability committee, every cross-chain messaging protocol, and every institutional tokenization stack with a transfer-agent or compliance gate.

Lux Industries has further filed a coordinated family of seven post-quantum cryptography patent applications extending the original architecture into the post-quantum era. The combined IP estate covers the architecture across classical, hybrid, and NIST-standardized post-quantum cryptographic primitives.

This document is a proposal to BlackRock for a Tier-2 institutional-tokenization license under the Lux IP estate. The license provides BlackRock with (i) operational certainty for the current BUIDL infrastructure and the announced tokenization-fund expansion; (ii) a competitive moat in institutional tokenized finance via first-mover field-of-use positioning; (iii) cryptographic and regulatory cover for the post-quantum migration that NIST and the OCC will require over the coming five to ten years; and (iv) cross-license backstop on BlackRock’s own digital-asset patent portfolio. We propose a deal structure consisting of a lump-sum license fee plus a capped per-fund running royalty, closing within sixty days from the date of substantive engagement.

The remainder of this document explains the patent estate, BlackRock’s specific exposure surface, the proposed license terms, indicative economics, and the recommended next steps for engagement.

1 The patent estate

1.1 U.S. Patent 11,803,853 B2 — the foundational asset

The Nahmii patent, granted 31 October 2023 with a priority date of 4 May 2018, claims an architecture for controlling execution of blockchain transactions in which the system is decomposed into five functional components: (i) a first set of participant nodes generating transactions; (ii) a server originating transactional records on behalf of users; (iii) a second set of nodes independently validating the records for completeness (data availability) and correctness (fraud); (iv) a second agent obtaining validation results from the second set of nodes; and (v) a first agent comprising a smart contract on a parent blockchain that permits or denies execution based on the validation result. A threshold number of validating nodes is required to substantiate completeness or correctness.

This architecture is the structural foundation of the modern blockchain scaling and cross-chain interoperability landscape. It reads on:

- **Optimistic rollups:** Optimism, Coinbase Base, Arbitrum One, and the broader OP Stack and Arbitrum Orbit ecosystems.
- **Alternative-DA rollups:** Mantle, OP Stack chains using EigenDA, Celestia, Avail, or AWS S3 as the data-availability layer; every Arbitrum Orbit AnyTrust deployment with a Data-Availability Committee.
- **Cross-chain messaging:** Chainlink Cross-Chain Interoperability Protocol (CCIP) with its Commit DON, Executing DON, and Risk Management Network; LayerZero V2 with Decentralized Verifier Networks and Executors; Axelar General Message Passing with validator-approved gateway execution; Wormhole guardian-attested message delivery; Hyperlane Interchain Security Module.
- **Rollups-as-a-Service platforms:** Alchemy Rollups, AltLayer, Caldera, Conduit, Gelato Rollups — where Alt-DA or AnyTrust configurations are operated for customer chains.
- **Institutional tokenization:** regulated transfer-agent, broker-dealer, and alternative-trading-system stacks where mint, burn, transfer, and settlement operations are gated by a transfer-agent + custody-MPC + compliance-gate three-party authorization architecture.

The patent is enforceable through 6 May 2039.

1.2 Post-quantum continuation family

Lux Industries filed a coordinated family of seven post-quantum patent applications in late 2025 to 2026. The family consists of:

Application	Subject matter
Master Continuation-in-Part of 11,803,853	Post-quantum-authenticated validation in operator-action systems. Inherits the 2018 priority date for an in the parent; the post-quantum-specific matter claim date.
PQ Data-Availability Attestation (provisional)	Threshold post-quantum attestation for Layer-2 and availability committees. Targets Alt-DA / DAC / Aments.
PQ Cross-Chain Message Gating (provisional)	Threshold post-quantum verifier-network attestation message delivery. Targets LayerZero, CCIP, Axelar, perlane future PQ migrations.
PQ Institutional Tokenization Settlement (provisional)	Transfer-agent + custody-MPC + compliance-gate thorization with post-quantum threshold signature ulated tokenization stacks (BUIDL-class funds, to market funds, tokenized treasuries, regulated stableco
Cryptographically Agile Validation Layer (provisional)	Epoch-based primitive rotation with quantum-threat regulatory-directive policy import, and graceful de cryptanalysis advances. Covers the algorithm-agility any PQ migration requires.
PQ Threshold MPC for Regulated Custody (provisional)	Cryptographically-enforced regulated custody: tran as a cryptographic constituent of the threshold rath dural prerequisite. Targets institutional digital-asset

The combined estate covers the architecture across every NIST-standardized post-quantum primitive (ML-DSA per FIPS 204; SLH-DSA per FIPS 205; ML-KEM per FIPS 203; HQC per the 2025 additional KEM selection) and across hybrid classical-and-post-quantum constructions. The estate is therefore primitive-agnostic and architecture-agnostic: every cryptographic migration path that brings post-quantum security into a blockchain transaction-control system enters the Lux IP zone.

1.3 Why this estate is uniquely durable

Three structural properties make the estate unusually durable as an institutional licensing asset.

Priority date precedence. The 2018 priority date predates every modern rollup architecture in production today (Optimism Bedrock 2023; Cannon fault proofs 2024; Arbitrum BoLD 2024; CCIP mainnet 2023; LayerZero V2 2024; Base 2023). Defendants cannot escape via earlier-priority prior art owned by themselves.

Plasma-prior-art-aware claim construction. The strongest pre-2018 prior art (Plasma 2017; Minimal Viable Plasma 2018) describes earlier embodiments of the same general pattern. Lux’s claim construction strategy distinguishes via threshold formalization, completeness/correctness dichotomy, and the two-agent (first-agent + second-agent) architecture. Continuation strategy provides a backstop: even if any parent claim is narrowed in inter partes review, continuations preserve the priority date with adjacent-scope claims.

Post-quantum extension. The PQ family closes the future-architecture escape route. Any operator migrating to post-quantum cryptography in response to regulatory mandate enters the patent zone regardless of which primitive is chosen.

2 BlackRock’s exposure surface

This section maps each of BlackRock’s current and announced tokenized-finance activities against the patent claims. The analysis is provided for executive context and does not constitute formal infringement assertion.

2.1 BUIDL — USD Institutional Digital Liquidity Fund

The BUIDL fund operates a tokenized money-market vehicle issued on multiple Layer-1 blockchains and accessed by institutional investors through Securitize as the registered transfer agent. As tokenized-fund AUM scales (BUIDL crossed \$1.7B in 2024 with announced expansion across additional Layer-1 chains), the fund’s operating architecture increasingly depends on:

- Cross-chain availability via interoperability protocols (CCIP, LayerZero, or bridge alternatives).
- Compliance-gated mint, burn, and transfer per the Securitize transfer-agent authority.
- Custody under Coinbase Institutional or a comparable regulated custodian using threshold MPC signing.

- Settlement gating by on-chain compliance checks tied to investor accreditation status.

Each of these architectural surfaces reads against U.S. Patent 11,803,853, with the compliance-gated mint/burn/transfer surface being a particularly clean read (Securitize as the transfer-agent “server”; the Securitize compliance gate as the “second set of nodes” validating the proposed operation; the BUIDL token contract as the “first agent” permitting or denying based on validation; the compliance-claim verification as the “second agent”).

Anticipated migration of BUIDL operations to dedicated Layer-2 or appchain infrastructure (publicly discussed at the 2025 institutional-digital-assets conferences) brings the rollup-specific claim elements into scope.

2.2 IBIT — iShares Bitcoin Trust

IBIT custody operations rely on Coinbase Institutional’s cold-storage and qualified-custody offering. The custody architecture includes threshold-signing arrangements that, when migrated to post-quantum primitives over the coming five-to-ten-year horizon, fall within the PQ Threshold MPC for Regulated Custody application. The future-state architecture also intersects with the Cryptographically Agile Validation Layer application during the migration window.

2.3 Announced future tokenized funds

BlackRock has publicly indicated expansion of the tokenized-fund product line beyond BUIDL into additional asset classes (tokenized Treasuries, tokenized money-market variants, tokenized real-world assets). Each expansion replicates the regulated-tokenization architecture and therefore the same surface against U.S. Patent 11,803,853 and the PQ Institutional Tokenization Settlement application.

2.4 Cross-chain infrastructure dependencies

BlackRock’s current and announced tokenization activities depend on cross-chain messaging infrastructure operated by Chainlink Labs (CCIP), LayerZero Labs (LayerZero V2), or Axelar Network. Each of these protocols is in the patent zone for U.S. Patent 11,803,853 and the PQ Cross-Chain Message Gating application. Should any of these protocol operators face license obligations or operational restrictions, BlackRock’s cross-chain availability for institutional flows is at risk. A BlackRock Tier-2 license includes protections against this scenario through covenants flowing from the underlying protocol licenses Lux Industries grants.

2.5 Post-quantum migration mandate

The Office of the Comptroller of the Currency, the Federal Reserve, the European Central Bank, and equivalent prudential regulators have begun issuing post-quantum cryptographic guidance for the financial sector. NIST’s 2024 finalization of FIPS 203 / 204 / 205 and the 2025 HQC selection initiate a mandatory migration on a 2027–2032 horizon for institutional infrastructure with data-protection horizons measured in decades. Tokenized-fund operating infrastructure, custody arrangements, and cross-chain messaging will all require post-quantum migration. Each migration step enters one of the Lux PQ applications.

3 The proposed Tier-2 institutional-tokenization license

The proposed license provides BlackRock with comprehensive coverage across the patent estate, structured for institutional reporting and operational predictability.

3.1 Scope

The license grants BlackRock a non-exclusive but field-restricted right to practice the entire Lux IP estate (U.S. Patent 11,803,853, the post-quantum patent family, and any continuations or divisions issuing therefrom) for the purpose of operating tokenized regulated financial instruments within the field of institutional asset management.

The field of use is defined as: (i) issuance, custody, settlement, and transfer of regulated digital securities and regulated digital fund instruments; (ii) cross-chain availability of such instruments via licensed protocols; (iii) post-quantum-cryptographic migration of (i) and (ii); and (iv) custody arrangements ancillary to (i)–(iii).

Activities outside the field of use — for example, operating a public consumer-facing rollup or general-purpose cross-chain messaging protocol — are excluded from the grant and require separate licensing.

3.2 Term

The license term runs from execution through the expiration of the last patent in the estate (currently 2039 for the original Nahmii patent; later for the post-quantum family depending on prosecution timing). The license is non-cancellable except for material breach.

3.3 Field-of-use exclusivity (most-favored-licensee)

Within the defined institutional asset management field, Lux grants BlackRock a most-favored-licensee right. If Lux subsequently licenses any other institutional asset manager (Vanguard, Fidelity, State Street, BNY Mellon, JPMorgan Asset Management, or comparable) on terms more favorable than BlackRock’s, those more-favorable terms flow to BlackRock automatically without renegotiation.

Lux retains the right to license institutional asset managers on terms equal to or less favorable than BlackRock’s, and retains the right to license outside the institutional asset management field (for example, to Coinbase as a custodian, or to Chainlink as a protocol operator) without restriction.

3.4 Sub-licensing

BlackRock receives a have-made right to sublicense the field-of-use rights to its appointed agents acting on its behalf within the field: registered transfer agents acting for BlackRock-sponsored funds, qualified custodians providing custody to BlackRock-sponsored funds, and named technology providers operating infrastructure on BlackRock’s behalf. The sub-license rights do not extend to third parties operating outside this principal-agent relationship.

3.5 Cross-license back

BlackRock grants Lux Industries a non-exclusive cross-license to BlackRock’s currently-issued and future digital-asset patent portfolio within the field of distributed-ledger-technology platform in-

frastructure. The cross-license back protects Lux against future BlackRock IP assertion and accumulates portfolio strength for the Lux ecosystem.

3.6 No-challenge covenant

BlackRock covenants not to challenge the validity or enforceability of any patent in the Lux estate for the term of the license, and not to support such challenges by third parties through expert testimony, financial contribution, or other material assistance. This is a standard institutional-license clause structured to be Lear-compliant under U.S. Supreme Court precedent.

3.7 Defensive termination

If BlackRock asserts any patent against Lux Industries or against any other licensee of Lux’s IP estate, the license terminates and any prepaid fees are forfeited. This clause is the institutional-license equivalent of a non-aggression pact.

4 Indicative economics

We provide indicative pricing for an institutional Tier-2 license. Final terms are subject to negotiation; the numbers below establish the order of magnitude and the structure.

4.1 Lump-sum component

A signing-date lump-sum license fee in the range of \$80M to \$150M, depending on the breadth of the field-of-use definition and the scope of the BlackRock cross-license back. The lump-sum covers (i) the present-state operation of BUIDL and IBIT through the current calendar year; (ii) prepaid running royalty for the first three years of operation in the expanded tokenized-fund product line; and (iii) the post-quantum-migration license under the PQ family.

4.2 Running royalty

A running royalty of 0.50% of gross fee revenue from BlackRock-sponsored tokenized regulated funds operating on-chain, payable quarterly. The royalty is capped at \$25M per calendar year, providing BlackRock with predictable maximum exposure as tokenized AUM scales.

The royalty base includes fund management fees, custody fees, and transaction fees collected by BlackRock and its sub-licensed agents from BlackRock-sponsored tokenized funds. The royalty base excludes (i) revenue from non-tokenized BlackRock operations; (ii) revenue from tokenized funds operating in jurisdictions outside the field-of-use definition; and (iii) revenue from BlackRock-managed non-fund accounts (separately managed accounts, institutional advisory).

4.3 Estimated total economics

Under conservative tokenized-fund growth projections (BUIDL plus three additional tokenized money-market or tokenized-Treasury vehicles, scaling to \$15–25B aggregate AUM over five years), total license-attributable BlackRock cost over the first five years is in the range of \$200M to \$300M. This figure is in line with comparable institutional infrastructure licenses (compare: typical institutional core-banking platform licenses with multi-year subscription components, typical SWIFT message-format licenses for FX and cross-border activity).

The cost is recoverable by BlackRock from the competitive moat created by exclusive-in-field positioning. Institutional tokenization competitors who do not hold the Lux license face open-ended IP exposure on their tokenized-fund operations, providing BlackRock with a durable competitive advantage during the period of greatest institutional-tokenization growth.

5 Strategic rationale for BlackRock

We summarize the strategic value of the license to BlackRock under five headings.

5.1 Operational certainty

The license eliminates IP-overhang risk for the BUIDL fund and announced future tokenized-fund expansion. BlackRock’s product roadmap is no longer subject to potential interruption from third-party patent assertion or from infrastructure-protocol licensing disputes. Auditors, regulators, and institutional limited partners can be told with full confidence that the platform IP foundation is owned, licensed, and durable through 2039 and beyond.

5.2 Competitive moat via first-mover field-of-use positioning

Lux is in advanced discussions with multiple institutional asset managers regarding the Tier-2 institutional license. Closing first establishes BlackRock as the institutional reference licensee, sets the comparable for subsequent licensees, and provides field-of-use most-favored-licensee protection against later entrants receiving better terms. The field-of-use definition can be calibrated to provide BlackRock with structural advantage in the institutional tokenization market relative to non-licensed competitors.

5.3 Post-quantum readiness

The license includes the PQ patent family. As the OCC, the Federal Reserve, the ECB, and equivalent prudential regulators issue post-quantum migration mandates over the coming five-to-ten-year horizon, BlackRock will need IP-clean migration paths for BUIDL infrastructure, IBIT custody, and the broader tokenized-fund product line. The Tier-2 license provides this in a single arrangement, avoiding the more difficult negotiation BlackRock would face if licensing the PQ family separately at a later date.

5.4 Cross-chain availability protection

The license includes the protocol-licensing flow: should Chainlink Labs, LayerZero Labs, Axelar Network, or other cross-chain infrastructure operators face IP disputes with Lux Industries that affect their availability to BlackRock, the BlackRock license includes covenants ensuring continued cross-chain availability for BlackRock-sponsored tokenized funds.

5.5 Cross-license backstop on BlackRock IP

The cross-license back protects BlackRock against future Lux IP assertion in adjacent fields and provides Lux with portfolio-strength contributions that benefit the broader BlackRock ecosystem through ecosystem-licensee discounts and field-of-use coordination.

6 Comparison: Tier-2 license versus other paths

For BlackRock, three paths exist with respect to the Lux IP estate.

Path	Strategic implication	Expected cost
Tier-2 license now	Operational certainty, first-mover field-of-use, included PQ family, included protocol-licensing flow. Closes within 60 days.	\$200M–\$300M over 5 years (lump sum + capped royalty)
Defer engagement	Continued IP-overhang risk. Subsequent license available but at later terms; field-of-use exclusivity may be unavailable if a competitor licenses first.	Higher 5-year total cost (\$300M–\$500M estimated) plus opportunity cost of lost first-mover positioning
Defensive engagement	Engage only after assertion or after market signal indicates assertion intent. Possibility of inter partes review challenge.	High legal cost (\$10M–\$30M defending IPRs across the patent estate); uncertain outcome; loss of operational certainty during proceedings; reputational risk on BlackRock platform stability

The first path provides the lowest total cost and the highest strategic value. The second path is acceptable if BlackRock places low value on field-of-use exclusivity and accepts the operational uncertainty. The third path is not commercially rational given the size of BlackRock’s tokenization commitments and the durability of the patent estate.

7 Recommended next steps

We propose the following sequence for engagement, structured to close within sixty days from substantive engagement.

- Step 1. Initial meeting** between Lux Industries leadership and BlackRock Financial Markets Advisory or BlackRock Digital Asset Strategy. Approximately ninety minutes. Agenda: review of the patent estate, BlackRock exposure analysis, license framework. Confidentiality agreement in advance.
- Step 2. Term sheet exchange** within fourteen days of initial meeting. Lux Industries provides the proposed Tier-2 term sheet. BlackRock provides comments and counterproposals on field-of-use, royalty structure, and cross-license back scope.
- Step 3. Diligence access**, both directions. Lux Industries provides BlackRock counsel with patent prosecution histories, claim charts, and prior-art memorandum. BlackRock provides Lux Industries with the BlackRock digital-asset patent portfolio relevant to the cross-license back.
- Step 4. Definitive agreement** drafted by BlackRock outside counsel using the agreed term sheet as the structural foundation. Lux Industries reviews and counter-marks within two weeks.

Step 5. Signing and announcement. Joint or unilateral announcement at BlackRock’s discretion. Lux Industries does not require public announcement; confidentiality of the license terms is fully acceptable.

8 Contact

Direct engagement contact at Lux Industries:

- Zach Kelling, CEO, Lux Industries, Inc. — `z@lux.industries`
- General counsel: `counsel@lux.industries`

Acknowledgement requested within thirty days of receipt of this document. Lux Industries is in parallel discussions with other institutional asset managers; the field-of-use first-mover positioning available to BlackRock depends on closing-first timing.

Confidentiality

This document is confidential and contains material non-public information regarding Lux Industries’ intellectual property estate, licensing strategy, and commercial positioning. Recipients may share within BlackRock on a need-to-know basis subject to standard confidentiality controls. External distribution requires prior written consent of Lux Industries.

Reservation of rights

This document is a proposal and is not a binding offer. Lux Industries reserves all rights with respect to the patent estate, including the right to assert the patents against any party operating outside an executed license arrangement. Nothing herein constitutes a covenant not to sue, a license grant, or a waiver of any patent rights.